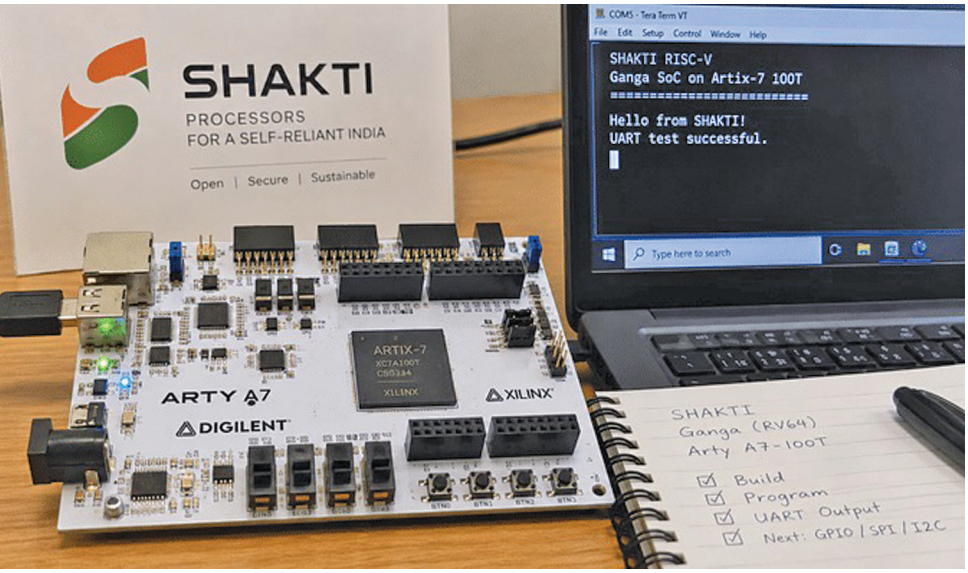
FPGA board runs homegrown RISC-V



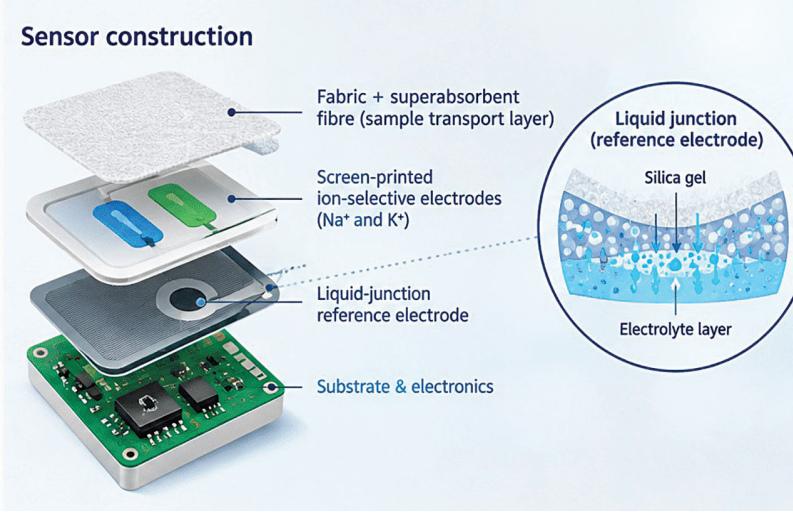
Three teams won the Digital India RISC-V Grand Challenge using SHAKTI and VEGA processor platforms, judged at SEMICON India 2026 (Credit: https://shakti.org.in)

India's 64-bit SHAKTI RISC-V processor can now run as a soft SoC on Artix-7 100T FPGA boards, giving developers a practical platform to test the indigenous architecture. Developed by the RISE group at IIT Madras, the Ganga implementation uses a six-stage in-order C-class core with RV64IMAC support, 256MB DDR, and 8KB boot ROM. Its peripherals include GPIO, SPI, UART, and timers, enabling embedded control, sensor interfacing, and hardware-software co-design. The FPGA-based implementation lets developers modify and rebuild processor configurations without fabricated silicon, supporting RISC-V and Indian processor experimentation.

Wearable sensor tracks sweat electrolytes

A wristwatch-type wearable sensor developed at Tokyo University of Science, Japan, tracks sodium and potassium signals in sweat in real time, potentially improving hydration monitoring for workers and athletes in hot environments. It combines screen-printed ion-selective electrodes, a rapid-stabilising reference electrode, and wireless electronics for data transmission and local storage. Tested on a construction worker during routine activities, the prototype recorded signals for about 2 hours 45 minutes. Further validation is needed to establish reliable electrolyte concentration measurements before practical health and occupational safety applications.





Wearable sensor tracks sweat electrolytes (Credit: www.industryevents.com)